

VISHAL KODAM

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Summary

Embedded and platform validation engineer with experience in firmware validation, board bring-up, hardware-software integration, automation scripting, and system-level debugging across Linux and Windows environments. Hands-on experience validating microcontroller-based and x86 platforms, developing Python/C/C++ automation scripts, working with Qt-based validation utilities, and debugging issues using oscilloscopes, logic analyzers, JTAG, serial logs, and protocol traces. Strong foundation in digital and analog electronics, communication interfaces, firmware behavior, and structured root-cause analysis.

Technical Skills

Testing & Validation: Firmware Validation, Platform Validation, System-Level Testing, Regression Testing, Functional Testing, Stability Testing, Power Validation, Hardware Triage, Root-Cause Analysis

Automation & Validation Tools: Python Automation Scripts, C/C++ Test Utilities, Qt-Based Validation Tools, Log Parsing Tools, Serial Console Tools, Test Execution Scripts, Debug Data Collection

Embedded & Platform Interfaces: SPI, I2C, UART, GPIO, USB, PCIe, RS232, RS485, TCP/IP, DAC Fundamentals

Firmware & Systems: Embedded C/C++, Bare-Metal Concepts, Firmware Validation, Driver/Firmware Interaction, BIOS/UEFI Validation Exposure, Embedded Linux, x86 Platform Fundamentals

Debug & Lab Tools: Oscilloscope, Logic Analyzer, Protocol Analyzer, JTAG, J-Link, FTDI, WinDbg, Firmware Flashing Tools

Hardware Fundamentals: Digital and Analog Circuits, Signal Timing, Signal Integrity Awareness, EMI Awareness, Power Distribution, Sensor Interfacing

Languages & OS: C, C++, Python, PERL, Bash, Linux, Windows, Ubuntu

Professional Experience

Google (NCV Telecom)

Jun 2025 – Present

Quality Engineer – Embedded / Platform Validation

Omaha, Nebraska

- Develop and execute firmware and embedded system validation tests for controller-driven platforms, focusing on interface behavior, hardware communication, and system stability.
- Create and maintain Python automation scripts to execute validation workflows, collect logs, parse test results, and reduce manual effort during regression testing.
- Develop C/C++ test utilities to exercise firmware interfaces, validate controller responses, and support repeatable hardware-level test execution.
- Work with Qt-based validation and debug utilities to configure test parameters, monitor execution status, capture logs, and analyze firmware or platform behavior during validation.
- Validate communication across SPI, I2C, UART, GPIO, and USB interfaces by checking protocol behavior, signal timing, and firmware interaction with connected peripherals.
- Support board bring-up activities including firmware flashing, hardware configuration, peripheral initialization, serial console setup, and first-level hardware triage.
- Use oscilloscopes, logic analyzers, and protocol traces to debug timing issues, bus communication failures, signal irregularities, and controller-level defects.
- Investigate firmware, driver, and hardware interaction defects through log analysis, trace review, reproduction testing, and collaboration with firmware and hardware teams.
- Document validation results, failure signatures, debug observations, automation output, and reproduction steps to support defect tracking and release-readiness decisions.

SUNY Poly College of Health Sciences

Jan 2023 – Jan 2025

Graduate Research Assistant

Marcy, New York

- Developed C++ and Python-based data acquisition workflows for sensor-connected systems, supporting embedded-style testing and hardware-interfaced data collection.
- Built automation scripts to collect experimental data, organize test outputs, process signal measurements, and improve repeatability of lab validation workflows.
- Configured Linux-based automation environments to collect, process, and validate experimental data from hardware-connected research setups.
- Interfaced with sensors and embedded processing environments to capture signal behavior, validate data integrity, and troubleshoot communication or acquisition issues.
- Applied digital signal processing techniques to analyze sensor output, communication behavior, and system performance across experimental test conditions.

- Prepared technical reports summarizing test methodology, signal observations, validation results, automation workflows, and system-level performance findings.

Intel Corporation

May 2019 – Nov 2022

Platform Validation Engineer – Co-op

Bengaluru, India

- Supported early platform bring-up and validation for x86-based systems by testing firmware, BIOS/UEFI behavior, peripheral initialization, and hardware-software interaction.
- Executed platform validation tests across BIOS/UEFI, PCIe, USB, storage, and controller-based interfaces to verify functional readiness during system integration.
- Used automation scripts and validation utilities to run test sequences, capture logs, check pass/fail behavior, and support regression testing across platform builds.
- Assisted with firmware flashing, OS provisioning, board configuration, lab setup, and test environment preparation for validation and debug activities.
- Used oscilloscopes, logic analyzers, JTAG tools, serial logs, and system traces to debug electrical, firmware, and protocol-level issues.
- Performed signal timing, stability, and compatibility validation for digital interfaces and high-speed platform components.
- Reproduced and isolated system-level failures involving boot flow, device detection, peripheral communication, firmware initialization, and firmware-driver interaction.
- Collaborated with firmware, silicon, board design, and validation teams to investigate defects, validate fixes, and improve platform stability.

Education

State University of New York

Jan 2023 – Jan 2025

Master of Science in Computer Science

Marcy, New York

Indian Institute of Information Technology Nagpur

June 2018 – May 2022

Bachelor's in Electronics and Communication Engineering

Nagpur, India